

Basis of Statistics

Experimental Design - Master Degree in Biomedical Omics

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1 Introduction

2 Mathematical Background

- Probability and its Properties
- Probability Distributions
- Parameters in Probability

3 Seven Steps for a Proper Statistical Analysis

4 Extra - Some Other Data Analysis Techniques

- Regression
- Dimensionality Reduction and Clustering

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The systematic collection and arrangement of numerical facts or data of any kind; (also) the branch of science or mathematics concerned with the analysis and interpretation of numerical data and appropriate ways of gathering such data.

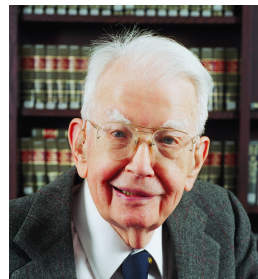
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if you torture the data long enough, it will confess

Ronald Coase [1910 - 2013]



Why do you Need Statistics?

Statistics is fundamental for experimental biologists [5]:

- extract useful information
- draw valid conclusions
- avoid common mistakes

[5] PMID: 31145670

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Unfortunately, statistical analysis often comes last in the lab, leading to the observation by the famous 20th century statistician R. A. Fisher

To consult [statistics] after an experiment is finished is often merely to [...] conduct a post mortem examination. [You] can perhaps say what the experiment died of.

[5] PMID: 31145670

Different Types of Statistics

■ Descriptive Statistics

- it's a *deductive method* (goes from the general to the particular)
- gives a summary of population data

■ Inferential Statistics

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Example. After collecting the scores of the statistics exams in a university for five years one can

- use **descriptive statistics** to obtain an overview of the scores in those years, and compare with the scores for the same exam in other universities;
- use **inferential statistics** to test hypothesis on the fact that five years ago the students studied more, or to predict the scores for the next year.

Population and Sample

Obviously, one would prefer to 'do statistics' on the whole world, in order to have the broader understanding. However, that is not feasible and we must settle for a limited amount of data.

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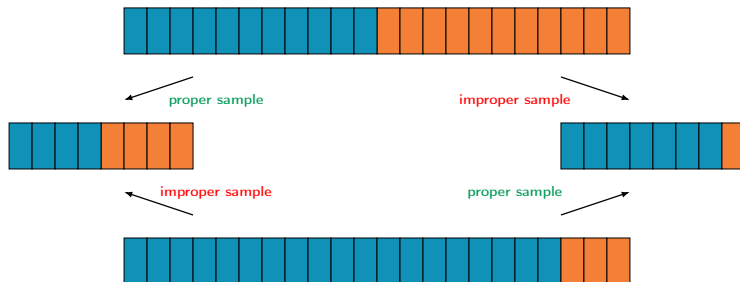
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The **population** is the group of all individuals with some common elements (e.g. people between 13 and 18 years)

The statistical **sample** is a subgroup of the population. We hope that this subgroup is representative of the whole population.



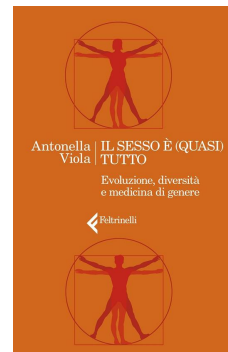
Importance of Proper Sampling

It is crucial for the sample to be *representative* for the population. Otherwise the inferred properties/informations will not describe the population.

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For instance, the symptoms for the Myocardial Infarction (MI) differs from male to female. However, most of the medical literature only reported the symptoms for males, probably because MI is much less fatal for female and less investigated.



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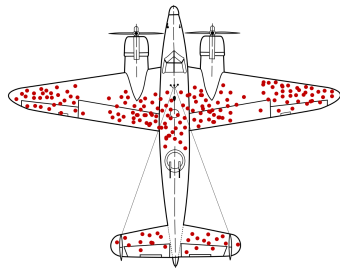
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- **Survivorship bias**: it is the tendency to concentrate to the entities that passed a selection process while ignoring or overlooking those who didn't pass. For instance, 100% of the people that played russian roulette say that it is safe.



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Probability

definition and first property

A *probability* is a “measure” of how likely is an **event** to happen.

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The sum of all mutually exclusive events sums to 1:

$$\Pr(\text{tail}) + \Pr(\text{head}) = 0.5 + 0.5 = 1$$

$$\Pr(\text{sunny day}) + \Pr(\text{rainy day}) + \Pr(\text{foggy day}) + \Pr(\text{cloudy day}) + \dots = 1$$



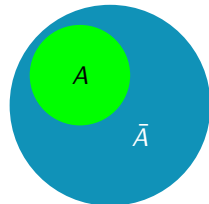
Probability

complement and multiple events

The *complement event* $\bar{A}$ of A is the event of **not** A

$$\Pr(\bar{A}) = 1 - \Pr(A)$$

$$\Pr(\text{die doesn't give 5}) = 1 - \Pr(\text{die gives 5})$$



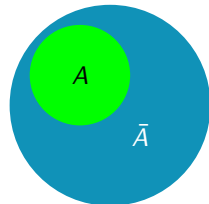
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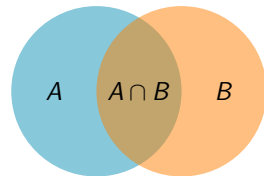


The probability of multiple events can be computed as follows

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

$\Pr(\text{die gives an even number or a multiple of three}) =$

$$\Pr(\{2, 4, 6\}) + \Pr(\{3, 6\}) - \Pr(\{6\})$$



Probability

conditional and independent events

The *conditional probability* $\Pr(A|B)$ of event A conditioned to B is the probability that A happens knowing that B already happened:

$\Pr(\text{a die gives 4} | \text{a die gave an even number})$

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Two events A, B are *independent* if the probability of A to occur does not affect the probability of B to occur

$$\Pr(A|B) = \Pr(A)$$

$\Pr(\text{a die gives 4 knowing that a coin toss resulted in tail}) = \Pr(\text{a die gives 4})$

Theorems

Theorem

For two events A and B,

$$\Pr(A \cap B) = \Pr(A|B) \Pr(B).$$

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Theorem

If two events A, B are independent, then

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What is a Distributions

In probability theory and statistics, a *probability distribution* is the **mathematical function**

$$p : E \rightarrow \mathbb{R}_{\geq 0}, \quad x \mapsto p(x)$$

that gives the probabilities of occurrence of different possible outcomes for an experiment.

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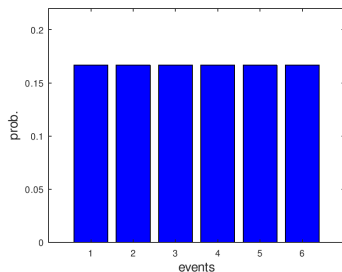


Figure: Probability distribution for a fair die

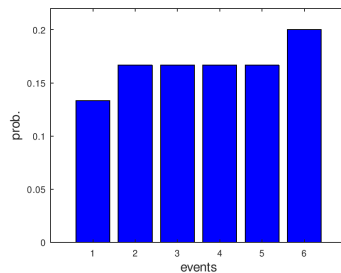


Figure: Probability distribution for an unfair die

Discrete and Continuous Distributions

what is a distribution

A distribution is **discrete** when the set of events E is discrete¹ (e.g. the result of a coin toss).

A distribution is **continuous** when the set of events E is continuous (e.g. the outcome of a measurement).

¹i.e. finite or, at most, computable

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Any function p with some properties is a valid probability distribution:

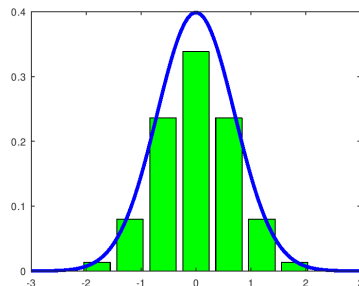
- the function is always positive:

$$p(x) \geq 0, \quad \forall x \in E$$

- the function “sums” to one

$$\sum_{x \in E} p(x) = 1$$

$$\int_E p(x) dx = 1$$



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Discrete and Continuous Distributions

what the distribution tells us

Be careful of what the probability distribution tells:

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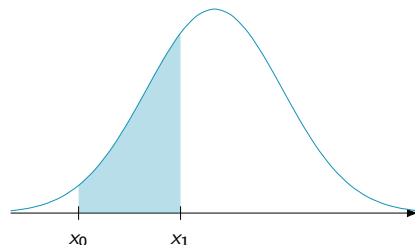
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- this is **not true** for a **continuous** distribution:
we can use the distribution to know the probability that a measurement is in a given interval:

$$\Pr(x_0 < x < x_1) = \int_{x_0}^{x_1} p(x) dx$$



Uniform Distribution

The uniform distribution **assigns at each event the same probability**.

It is both a **discrete** and a **continuous** distribution:

$$p(n) = \frac{1}{N}, \quad E = \{1, 2, 3, \dots, N\}$$

$$p(x) = \frac{1}{b-a}, \quad E = [a, b] = \{x : a \leq x \leq b\}$$

Example. The outcome of fair coins and dice are uniformly distributed.

Binomial (or Bernoulli) Distribution

The *binomial distribution* is a discrete distribution for events classifiable via **binary variable** (TRUE or FALSE).

Let's say that the probability for an event to occur (TRUE) is p , and the probability for the same event to not occur is $q = 1 - p$.

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In $n = 10$ die throws, the probability for a six to occur **exactly** $k = 3$ times is

$$P(3) = \frac{10!}{3!(10-3)!} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{10-3} \approx 0.155049 \dots$$

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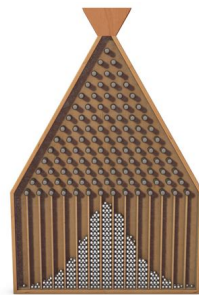
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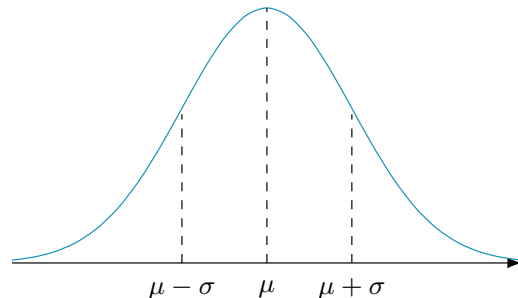
Galton Board

Normal (or Gaussian) Distribution

The normal distribution $\mathcal{N}(\mu, \sigma)$ is defined by two parameters: the *mean* μ and the *standard deviation* $\sigma > 0$ as

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- at $x = \mu$ the distribution reaches its peak;
- at $x = \mu \pm \sigma$ the function has its inflexion points (from convex to concave and vice-versa).



Normal (or Gaussian) Distribution

why is it so used?

The *normal distribution* is the most used and well-known distribution:

- it well describes the variability of measurement errors
- many phenomena in nature are well approximated by the normal distribution
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Theorem (Central Limit)

Let $\{X_1, X_2, \dots, X_n\}$ be a sequence of **independent** and **identically distributed** (i.i.d.) random variables. Then, the sample average

$$\hat{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

tends to be normally distributed as $n \rightarrow \infty$.^a

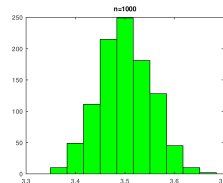
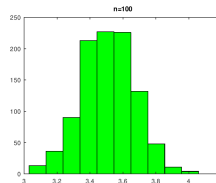
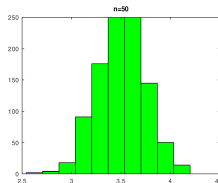
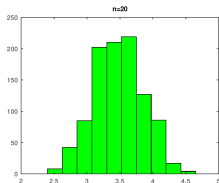
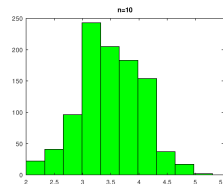
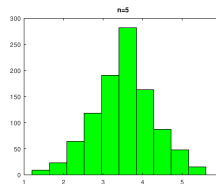
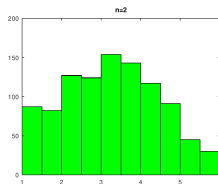
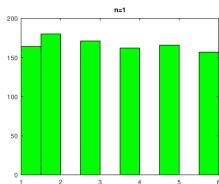
^a<https://tinyurl.com/2sb5thxa>



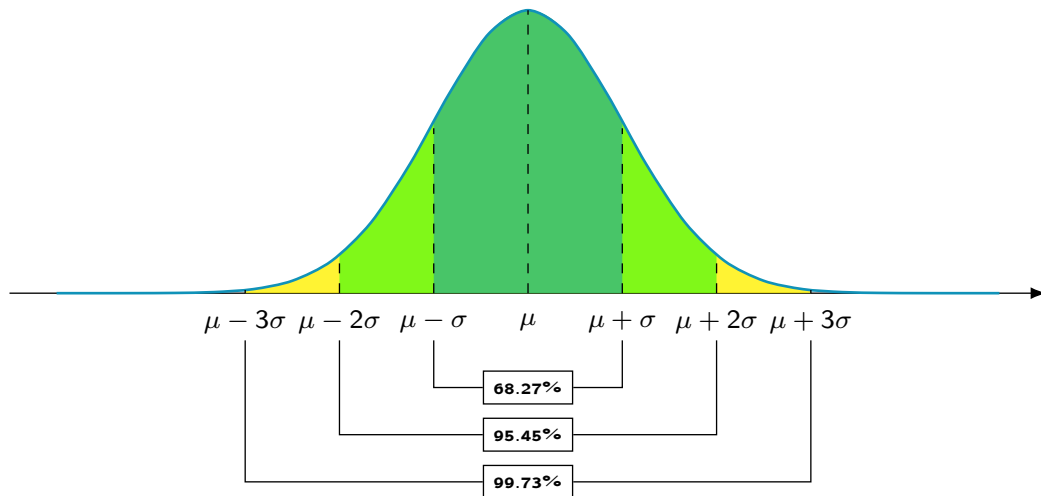
Central Limit Theorem

example

We simulate a die throw (i.e. a uniform distribution) and consider the sample average for different sample sizes n



Probability Under the Normal Distribution



Parameters in Probability

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- **dispersion** indices:

- variation range
- interquartile distance
- variance/standard deviation

- **shape** indices:

- symmetry
- kurtosis

Parameters in Probability

mean, median, and mode - definition

Suppose we sampled a set $\{X_1, X_2, \dots, X_n\}$ from an experiment. Then

Definition (Mean)

The *mean* is the average value, i.e. the sum divided by the amount of elements in the sample

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

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Definition (Median)

The *median* is the 'value in the middle' after sorting the sampled set.

Parameters in Probability

mean, median, and mode - example

Example. Let us suppose that we perform 11 die rolls that result in the following sample

$$\{4, 3, 6, 3, 2, 5, 1, 5, 6, 5, 5\}$$

Then the mean is

$$\frac{4 + 3 + 6 + 3 + 2 + 5 + 1 + 5 + 6 + 5 + 5}{11} = 4.\overline{09}$$

The mode is 5 since the die rolled a 5 a total of four times.

If we sort the sample, we see that the median is 5.

$$\{1, 2, 3, 3, 4, 5, 5, 5, 5, 6, 6\}$$

Parameters in Probability

variance

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$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}.$$

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Remark. As the sample size increases, the difference between the biased and unbiased variance decreases!

Parameters in Probability

standard deviation, coefficient of variation

Definition (Standard Deviation)

The *standard deviation* is the square root of the variance:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}, \quad S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

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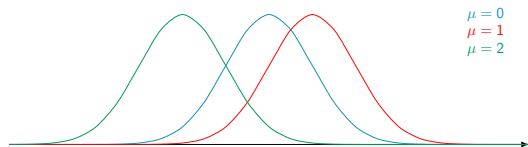
Definition (Coefficient of variation)

The *Coefficient of variation* (a.k.a. *relative standard deviation*) is the ratio between standard deviation and population (or sample) size

$$\text{C.o.V.} = \frac{\sigma}{N}, \quad \text{C.o.V.} = \frac{\sigma}{n}$$

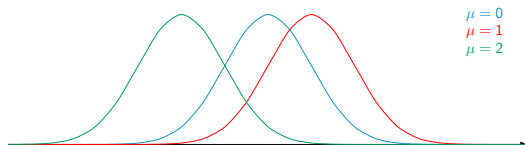
Mean and Standard Deviation in the Normal

A change in the mean μ causes a translation of the distribution.

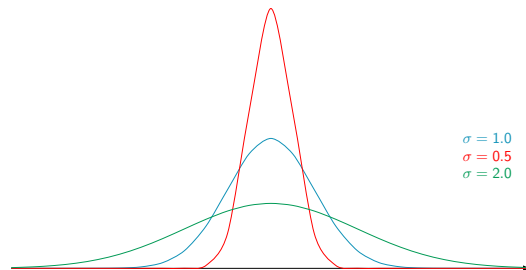


Mean and Standard Deviation in the Normal

A change in the mean μ causes a translation of the distribution.



A change in the standard deviation σ makes the distribution wider or narrower.



Parameters in Probability

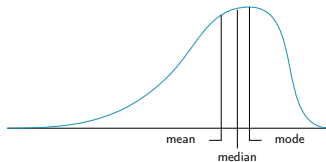
skewness

Definition (Skewness / third moment)

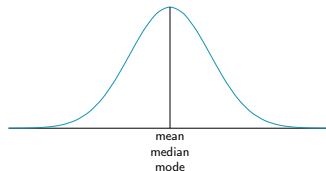
The *skewness* is defined as (resp. for population and sample)

$$\text{Skwn} = \frac{\sum_{i=1}^N (X_i - \mu)^3}{N \sigma^3}, \quad \text{Skwn} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{S} \right)^3$$

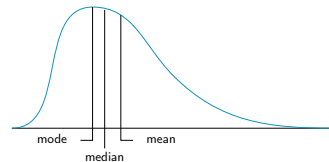
NEGATIVE SKEW



SYMMETRIC



POSITIVE SKEW



Parameters in Probability

kurtosis index

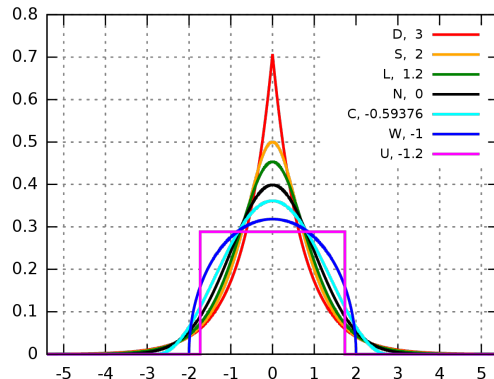
Definition (Kurtosis / fourth moment)

The *kurtosis* is defined as (resp. for population and sample)

$$\text{Kurt} = \frac{\sum_{i=1}^N (X_i - \mu)^4}{N \sigma^4} - 3$$

$$\text{Kurt} = \frac{n}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{S} \right)^4$$

- Kurt < 0 hyponormality (platikurtic curve)
- Kurt = 0 normality
- Kurt > 0 hypernormality (leptikurtic curve)



1 Introduction

2 Mathematical Background

- Probability and its Properties
- Probability Distributions
- Parameters in Probability

3 Seven Steps for a Proper Statistical Analysis

4 Extra - Some Other Data Analysis Techniques

- Regression
- Dimensionality Reduction and Clustering

Seven Steps

The paper *Empowering statistical methods for cellular and molecular biologists* [5] (PMID: 31145670) provides a list of seven steps to do a proper statistical analysis, from the experiment design to the data presentation.

- 1 Decide what you aim to estimate
- 2 Frame your biological and statistical hypotheses
- 3 Design your experiment
- 4 Examine your data and finalize the analysis plan
- 5 Perform a hypothesis test
- 6 Frame appropriate conclusions based on the statistical tests
- 7 Choose the best way to illustrate your results

Step 1: Define the Estimation Aim

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It should be mentioned that **in a population, the actual parameter has a fixed, unknown, value.** The goal is to **infer it from our sample.** For instance, in a population of cells, the division rate has a fixed mean and variance. However, we can only measure the division rate on a sample.

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This is why it is important to make **multiple measurements**, possibly with **different methods**:

With multiple measurements we **reduce the risk of improper sampling** and our estimate will be closer to the true population value.

Using different methods can help in **verify accurate methods and identify biased ones.**

Step 2: Frame your Biological and Statistical Hypotheses

A crucial step in designing a successful experiment is translating a biological hypothesis into **null** and **alternative statistical hypotheses**.

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In scientific research, the **null hypothesis H_0** is the claim that the differences between two set of data is due to chance alone. The **alternative hypothesis H_1** , instead, is the claim that differences between two set of data is consequence of the different experimental scenarios.

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For instance, an alternative hypothesis might be that the rate of degradation of *Epidermal Growth Factor Receptors* (EGFR) depends on the EGF dose in the medium. The null hypothesis would be that the ligand dose doesn't affect the degradation rate.

Step 3: Design your Experiment

nature and number of variables

All variables that could influence responses and are measurable should be recorded and considered in the statistical analysis, **not only those related to the intentional treatment** but also the *nuisance variables* (e.g. date of data collection, lot number of the reagent, room temperature and humidity, ...).

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It should be clear if your data will be *numerical* or *categorical*, since this determines which statistical tools you will need to analyze them:

- *numerical data* are measured as precise as possible and can have any numerical value. Typical numerical data are concentration, volume, temperature,
- *categorical data* are variables that can be chosen from a small set. Typical categorical data are gender, color, stages of cell cycle,

Warning: proportions and percentages are categorical responses even though they are numbers!

Step 3: Design your Experiment

methods of data acquisition

Common statistical tests **assume randomization and exchangeability**, meaning that all experimental units (e.g. cells) are equally likely to get each treatment and the data for one experimental unit is the same as that of any other receiving the same treatment.

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The challenge is to **understand your experiment well enough to randomize treatments effectively across potential confounding variables**. For instance, it is not wise to image all mutant cells one week and all control cells the next week, because differences in the conditions during the experiment could have confounding effects.

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Many statistical tests further assume that observations are *independent*. **When this is not the case** (e.g. repeated measurements of same specimen) **one should use methods that account for correlated observations**. Time-course studies are a typical example of experiment with repeated measurements.

Step 3: Design your Experiment

statistical test

Having decided the variables and the method to collect data, the next step is to select the appropriate statistical test.

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Most common statistical tests require predetermining an acceptable rate of *false positives* α and *true negatives* β .

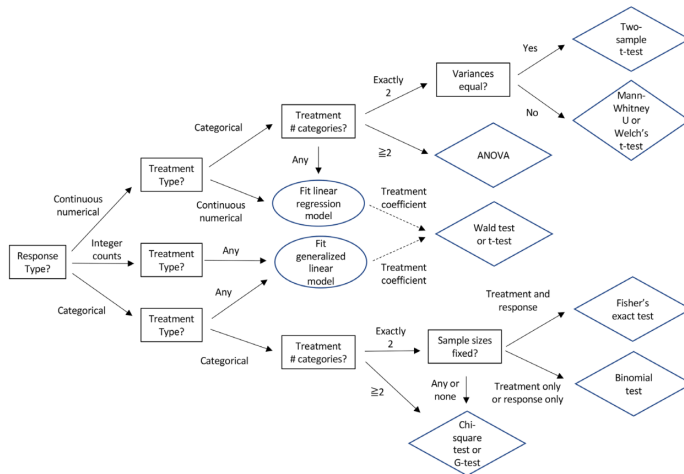
		null hypothesis H_0 is	
		true	false
decision about null hypothesis	accept	correct inference (true negative) $p = 1 - \alpha$	type II error (false negative) $p = \beta$
	reject	type I error (false positive) $p = \alpha$	correct inference (true positive) $p = 1 - \beta$



Remark: the null hypothesis is not being pregnant

Step 3: Design your Experiment

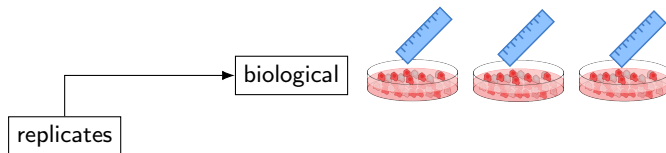
statistical test



Step 3: Design your Experiment

biological and technical replication

Biological replicates (i.e. measurements on separate samples) are used for parameters estimates and statistical tests, since they allow to **describe variation in the population**.



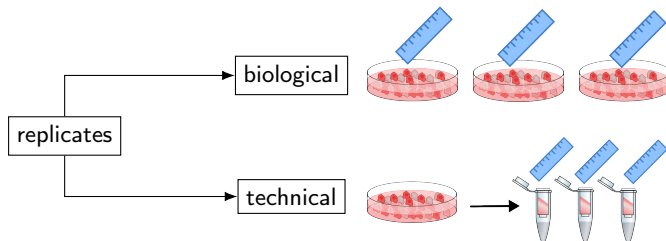
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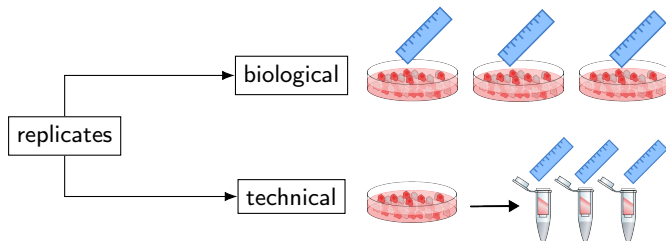
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treating technical replicates as biological ones is called *pseudoreplication* and often produces low estimates of variance and erroneous test results.



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Step 3: Design your Experiment

number of measurements

Extensive replication of experiments has multiple advantages:

- higher precision of parameter estimate;
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Luckily for us, [experimental biology has two major advantages over observational biology](#):

- experimental conditions are usually well controlled (using genetically identical organisms under precise laboratory conditions). This reduces the variation between samples and compensate to some extent the small sample size.
- the assignment of treatment to the specimens can be randomized in order to minimize the influence of confounding variables.

Step 3: Design your Experiment

statistical power

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Statistical power is **the probability of rejecting a truly false null hypothesis** and we want it as high as possible. Three variables contribute to it: **number of measurements**, **variability of those measurements** (i.e. standard deviation), and **effect size** (i.e. difference in response between control and treated populations)

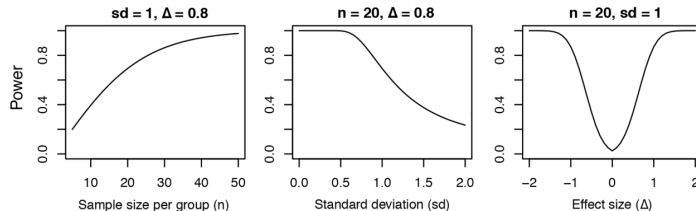
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Clearly, one does not know the outcome of an experiment in advance, but you can **estimate it from previous experiments**, or run some pilot experiments on the control sample to estimate the variability. Then, one can design the experiment knowing roughly how many measurements will be required.



Step 4: Examine the Data and Finalize the Analysis Plan

Experimental data **should not deviate strongly from the assumptions** of the chosen statistical test, and the sample sizes should be large enough to evaluate if this is the case. Strong deviations from expectations will result in inaccurate test results.

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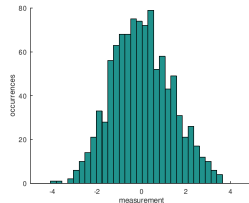
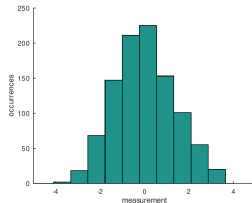
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even well designed experiments may require adjustments to the data analysis plan!

Most statistical tests are robust to small deviation from the “perfect” bell-shape curve of the normal distribution. Thus, a visual inspection of the measurements via a histogram is enough to decide if such a test will be fine.



Step 4: Examine the Data and Finalize the Analysis Plan

how to “normalize” data

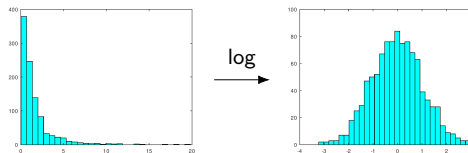
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For **right-skewed distributions**, you can take the logarithm, the square root, or the reciprocal of the measurements to ‘un-skew’ them.

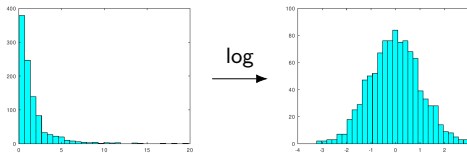


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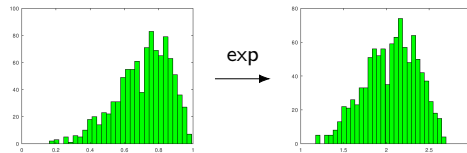
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For **left-skewed distributions**, you can take the exponential or the square of the measurements to ‘un-skew’ them.



Step 5: Perform a Hypothesis Test

A **hypothesis test** is performed to determine the probability of observing the experimental data if the **null hypothesis is true**.

To do so, these tests compare the properties of the experimental data with the theoretical distribution of outcomes expected when the null hypothesis is true.

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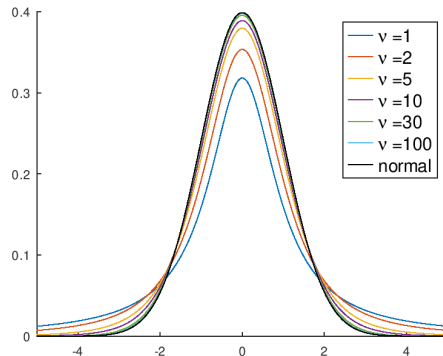
Example. for continuous measurements, the t -test compare the averages of two groups and determine whether the differences between them are more likely to arise from random chance (i.e. the null hypothesis). If the null hypothesis is true, the most likely outcome is no difference ($t = 0$).

By comparing the t -statistics from an experiment with the theoretical t -distribution, the t -test return the **probability that the experimental outcome occurred by chance**. If this probability, called p -value, is low, the null hypothesis is unlikely to be true.

Step 5: Perform a Hypothesis Test

The t -student distribution has one parameter, the *number of degrees of freedom* ν . In an experiment, the number of degrees of freedom is one less than the sample size.

This distribution (and the related test) is widely used because, as $\nu \rightarrow \infty$, the distribution tends to the normal distribution.



Step 5: Perform a Hypothesis Test

one-sample t -test

With one sample of size n , we test the **null hypothesis** that the true mean $\bar{x}$ is equal to the *null value* μ_0 :

$$t = \frac{\bar{x} - \mu_0}{\bar{\sigma} / \sqrt{n}},$$

where $\bar{\sigma}$ is the sample standard deviation.

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- if t is small, then either
 - the 'error' $\bar{x} - \mu_0$ is small
 - the 'noise' $\bar{\sigma}/\sqrt{n}$ is large relative to the error

and likely the population mean is the same as the null mean μ_0

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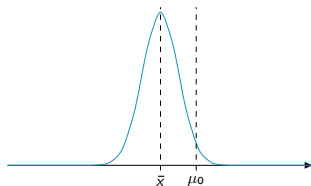
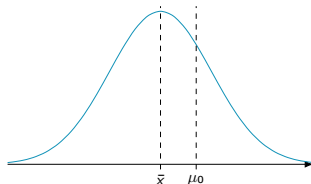
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Remark. The 'noise' is small either when the standard deviation within the sample is small, or when the sample is big.

Step 5: Perform a Hypothesis Test

one-sample t -test: the idea

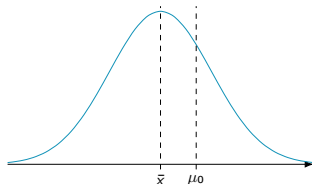


The null hypothesis is typically rejected either when

- the sample variance is small w.r.t. the error between μ_0 and $\bar{x}$
- the sample variance is not so small w.r.t. the error between μ_0 and $\bar{x}$ but a lot of samples were used

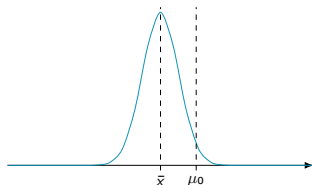
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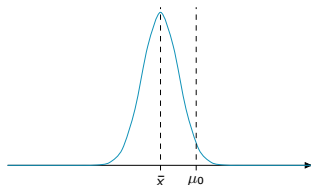
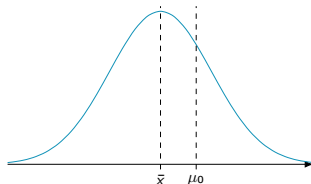


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The null hypothesis is typically accepted either when

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Basically, samples big enough make the estimate of the sample average enough precise to reject H_0 even with smaller error and bigger variances!

Step 5: Perform a Hypothesis Test

two-sample t -test

If an experiment comparing **two categorical treatments** (e.g. wild type and mutant cells) produces continuous measurements **normally distributed with equal variances**, the appropriate test is the *two sample t -test*, a.k.a. **Student's t -test**.²

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$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{s^2 (1/n_1 + 1/n_2)}}, \quad s^2 = \frac{\sum_{i=1}^{n_1} (x_1^{(i)} - \bar{x}_1)^2 + \sum_{j=1}^{n_2} (x_2^{(j)} - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

where $\{x_1^i : i = 1, 2, \dots, n_1\}, \{x_2^j : j = 1, 2, \dots, n_2\}$ are the samples.

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where $\{x_1^i : i = 1, 2, \dots, n_1\}, \{x_2^j : j = 1, 2, \dots, n_2\}$ are the samples.

Again, t is small when the two samples have similar averages and is large when the noise at the denominator is large w.r.t. the error at the numerator.

If t is small, it is likely that the two samples come from the same population and the null hypothesis is likely true (e.g. there are not differences between wild type and mutant).

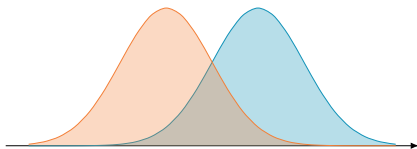
²'Student' is the pseudonym of William Sealy Gosset, the developer of the test

Step 5: Perform a Hypothesis Test

two-sample t -test: the idea

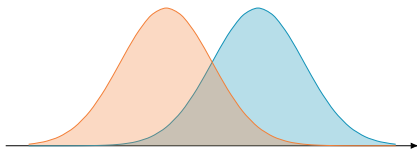
The null hypothesis is typically rejected either when

- the distributions for the two samples have a small overlap
- the distributions for the two samples have a significant overlap, but the samples are big enough



Step 5: Perform a Hypothesis Test

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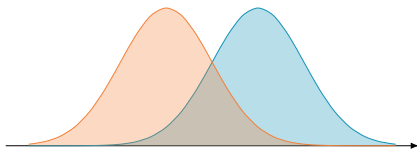
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Basically, **samples big enough make us reject the null hypothesis even when the distribution are not extremely different.** This because **with a bigger the sample size we are more sure about the estimated distribution.**

Step 5: Perform a Hypothesis Test

conversion of a t -statistic to a p -value

The statistical software converts the test statistic (e.g. the t from the Student t -test) into the so called p -value.

Definition (p -value)

The p -value is the probability of obtaining results *at least as extreme* as the observed results under the assumption that the null hypothesis H_0 is true.

Step 5: Perform a Hypothesis Test

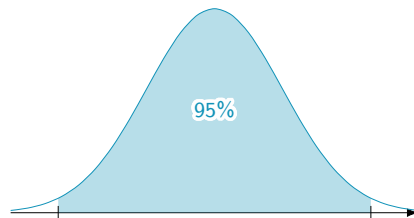
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For instance, if a p -value cutoff $\alpha = 0.05$ means that we accept a false-positive outcome in 5 out of 100 random trials. If the actual p -value is smaller or equal than the cutoff α , **the null hypothesis is rejected because the experimental data are unlikely under the null hypothesis H_0 .**



Step 5: Perform a Hypothesis Test

comparing the outcomes of multiple treatments

A common misconception is that a sequence of pairwise t -tests comparing each of several treatments and a control is equivalent to a single integrated statistical analysis (e.g. ANalysis Of VAriance, ANOVA). This is not true because **the sequence of pairwise tests is more vulnerable to false positive**, because the type I error rate is added across tests.

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Example. In an experiment with three treatments and a control the total type I error across the tests rises up to 0.26 with six pairwise t -tests each with cutoff $\alpha = 0.05$:

$$1 - (0.95)^6 = 0.264908 \dots$$

On the other hand, an ANOVA analysis test the null hypothesis that **all treatments and control have the same response** with $\alpha = 0.05$. If the test rejects H_0 , a post-analysis (e.g. via Tukey-Kramer) can determine which pairs differ significantly, while keeping the type I error rate below $\alpha = 0.05$.

Step 6: Frame Appropriate Conclusions Based on the Statistical Test

The fact that a statistical tests return a p -value below the cutoff **does not guarantee anything**: there is always a risk of having encountered a false positive! One should always **be skeptical and use judgment**.

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Some statisticians believe that the widespread use of $\alpha = 0.05$ has resulted in an **excess of false positive in biology**, suggesting smaller cutoffs [1]. Others even suggest to abandon tests of statistical significance altogether [4]. ^a

^a<https://tinyurl.com/nsej7asv>

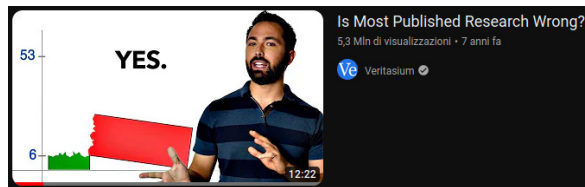


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Luckily, rigorous methods exist to determine whether **low statistical power** is the issue. Then, one should decide if it is appropriate to repeat the experiments or accept the result (and avoid wasting time, energy, and reagents).

Step 7: Choose the Best Way to Illustrate the Results

be precise and exhaustive

When presenting your results, the first thing to do is being clear and exhaustive on what was done: not only material and methods of the experiment, but also [type of statistical test and data pre/post-processing](#), as well as the [statistical software of choice](#) (R, python, Matlab).

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Specifying the software of choice is crucial to replicate tests results: different software may have different implementations of the same statistical test, and any difference in the implementation may cause an error.

	A	B
1	Formula	Result
2	$=1*(0.8-0.5-0.3)$	0.000000000000000055511151231258

Step 7: Choose the Best Way to Illustrate the Results

graph vs table

Depending on the type of data, one should decide to use a table instead of a figure.

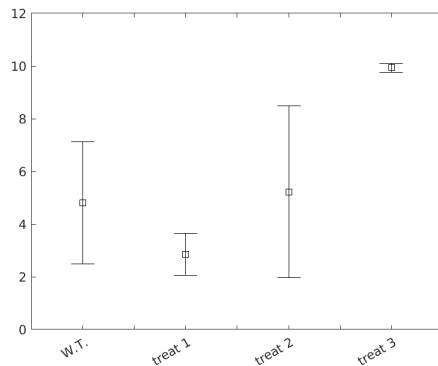
For instance, counts for a categorical treatment and categorical responses are well displayed by tables.

Step 7: Choose the Best Way to Illustrate the Results

create a proper graph

The first informations you should show in a figure are the **mean and standard deviation of your samples**.

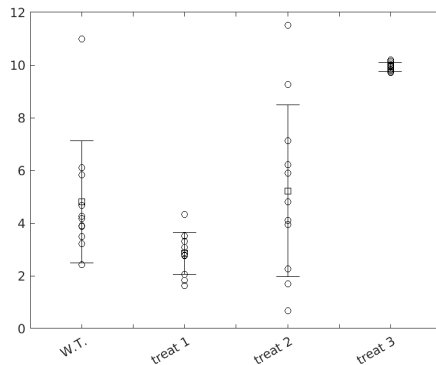
This is done via a plotting technique known in most softwares as **errorbar**.



Step 7: Choose the Best way to Illustrate the Results

create a proper graph

When possible, it is preferable to **show also the actual data points**.

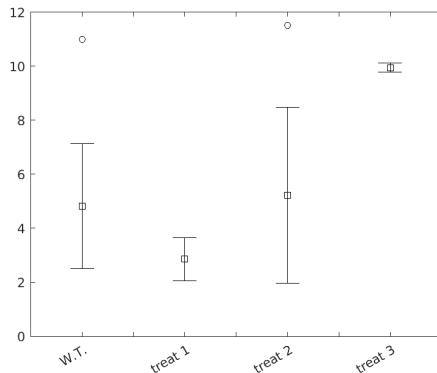


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create a proper graph

When possible, it is preferable to show also the actual data points.

Sometimes, especially when many data have been collected, one can limit themselves to show only the outliers.

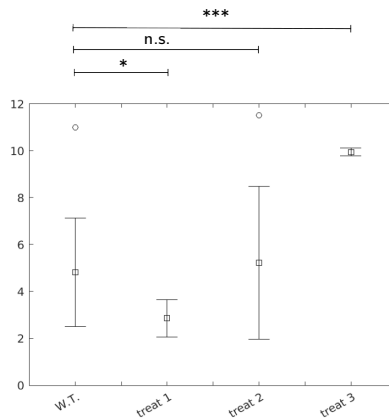


Step 7: Choose the Best Way to Illustrate the Results

create a proper graph

Finally, the result of the statistical test can be added to the plot.

n.s. - non significant
 * - $p < 0.05$
 ** - $p < 0.01$
 *** - $p < 0.005$

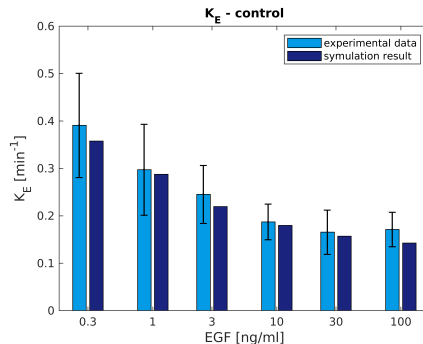
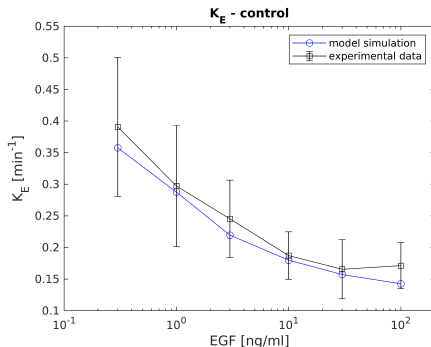


Step 7: Choose the Best Way to Illustrate the Results

experience and personal taste

It does not exist the perfect way to present your data!

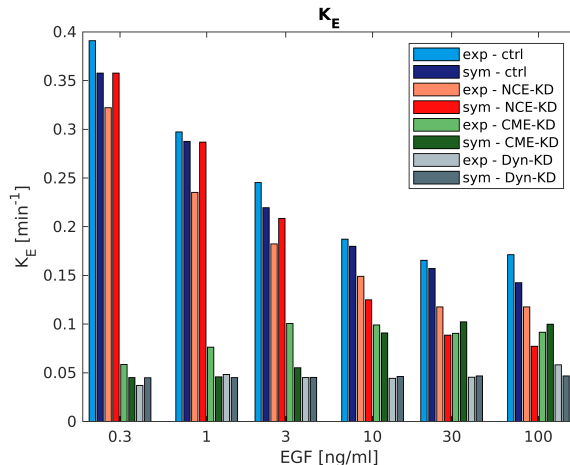
The same data can be presented in different ways, and experience (and personal taste) will guide you.



Step 7: Choose the Best Way to Illustrate the Results

experience and personal taste

When some plot stile are 'equivalent', one may be chosen for coherence when presenting the next results.



1 Introduction

2 Mathematical Background

- Probability and its Properties
- Probability Distributions
- Parameters in Probability

3 Seven Steps for a Proper Statistical Analysis

4 Extra - Some Other Data Analysis Techniques

- Regression
- Dimensionality Reduction and Clustering

Linear Regression

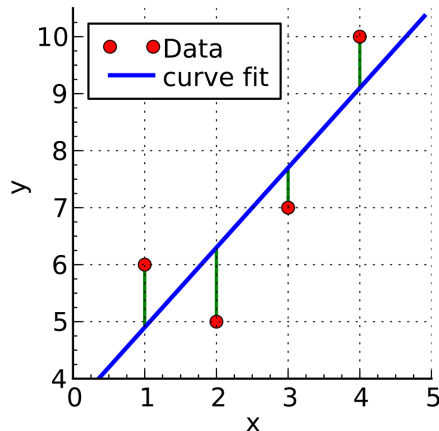
what is it?

Linear Regression is a technique to find (linear) relationship between two or more continuous variables:

$$y = \beta_0 + \beta_1 x.$$

Parameters β_0, β_1 are obtained by minimizing the **prediction error** of the experimental data

$$\varepsilon = \sum_{i=1}^N (\hat{y}_i - (\beta_0 + \beta_1 \hat{x}))^2.$$

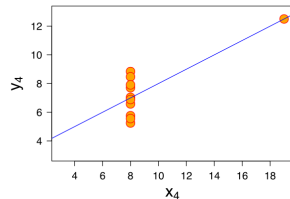
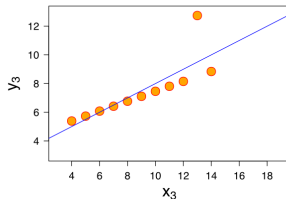
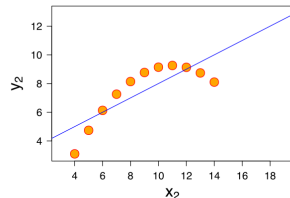
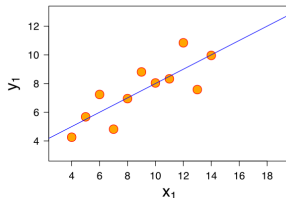


Linear Regression

be careful

As always, obtaining a linear model for the regression is not enough to guarantee that the variables are actually linearly dependent!

you should always be skeptical of what you discover



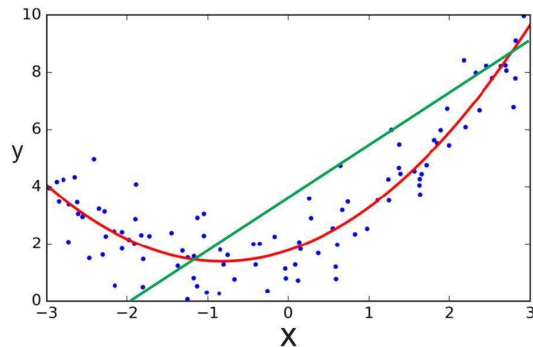
Polynomial Regression

Sometimes, a linear model may not be able to capture the relationship between data and you could decide to use a general polynomial:

$$y = \beta_0 + \beta_1x + \beta_2x^2;$$

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$$y = \beta_0 + \beta_1x + \beta_2x^2 + \beta_3x^3 + \dots;$$

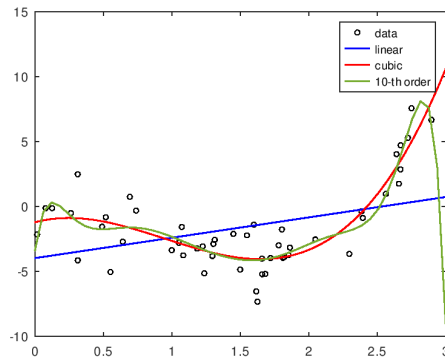


Polynomial Regression

overfitting

You should never use models that are too complex since they may result in **over-fitting**.

Over-fitting happens when the model is so complex that it does not describe simply the trend, but starts fitting also the noise of the data.



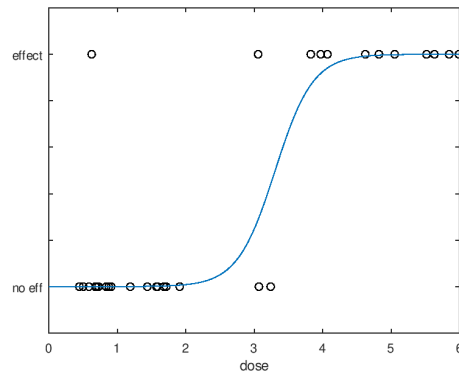
Logistic Regression

When the dependent variable is a binary variable (T/F), you should use *logistic regression*.

For instance, let's say that we want to determine the dose of a drug necessary to trigger an effect.

Then, the logistic regression gives a **probability of effect given the dose**, based on the logistic function

$$p(x) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x))}.$$



Principal Component Analysis

what is it?

Principal Component Analysis (PCA) is a technique for analyzing data with a high number of features. The **goal** of PCA is to reduce the dimensionality (i.e. the number of features) of the dataset.

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This is accomplished by linearly transforming the data into a new coordinate system where most of the variation in the data can be explained by few dimensions:

$$\hat{X} = \Omega X, \quad \hat{x}_n^{(i)} = \sum_{j=1}^F \omega_{ij} x_n^{(j)}, \quad i = 1, 2, \dots, m.$$

where

- $X \in \mathbb{R}^{F,N}$ is the dataset of N samples with F features;
- $\Omega \in \mathbb{R}^{m,F}$ is the transformation matrix from F to m features;
- $\hat{X} \in \mathbb{R}^{m,N}$ is the projected dataset of N samples with now m features;

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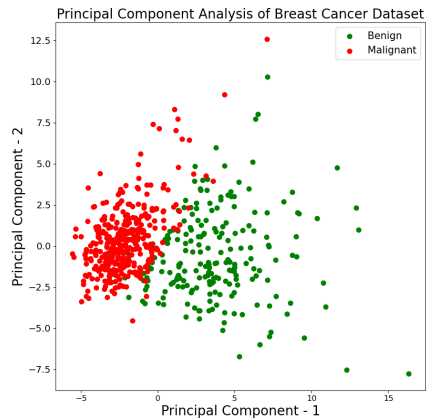
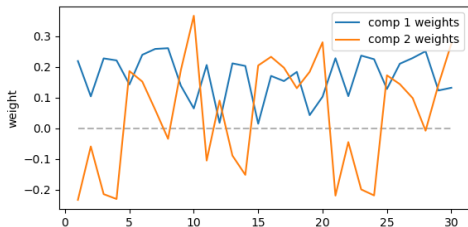
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It should always be clear that PCA **does not simply select the most important features**, but it **applies a (linear) transformation that combines all the features!**

Principal Component Analysis

example

The `breast_cancer` dataset contains 569 samples (212 malignant and 357 benign) with 30 features (radius, symmetry, smoothness, concavity, ...). After a PCA, we discover that two components are enough to explain 62% of the variance in the dataset (44% the first, 18% the second).

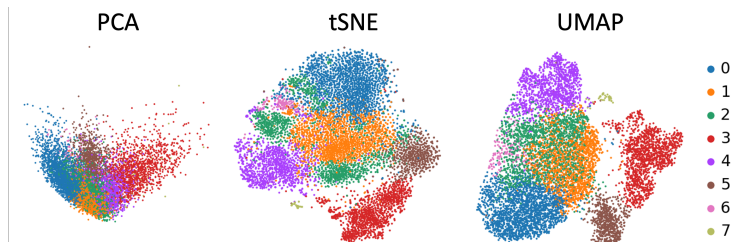


t-SNE and UMAP

PCA is a *linear* projection technique that uses covariances to define the projection map. However, in some cases, a linear projection cannot capture the variability in data. This is true especially in single cell analysis. In these cases other dimensionality reduction techniques are preferable.

For instance *t-distributed Stochastic neighbor Embedding* (t-SNE) use probability distribution between two any data-point that are higher when the two point are similar and try to put them in the low-dimensional space by minimizing the *Kullback-Leibler divergence*.

Uniform Manifold Approximation and Projection (UMAP), instead, tries to generate a small dimensional manifold on which the data are uniformly distributed.



References

- [1] Daniel J Benjamin et al. “Redefine statistical significance”. In: *Nature human behaviour* 2.1 (2018), pp. 6–10.
- [2] David Lane et al. *Introduction to statistics*. Citeseer, 2003.
- [3] Leland McInnes, John Healy, and James Melville. “Umap: Uniform manifold approximation and projection for dimension reduction”. In: *arXiv preprint arXiv:1802.03426* (2018).
- [4] Blakeley B McShane et al. “Abandon statistical significance”. In: *The American Statistician* 73.sup1 (2019), pp. 235–245.
- [5] Daniel A Pollard, Thomas D Pollard, and Katherine S Pollard. “Empowering statistical methods for cellular and molecular biologists”. In: *Molecular Biology of the Cell* 30.12 (2019), pp. 1359–1368.
- [6] Martin Wattenberg, Fernanda Viégas, and Ian Johnson. “How to use t-SNE effectively”. In: *Distill* 1.10 (2016), e2.

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[2] <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=a8182909e4e95c530f4849ba75f74c7c633651b0>

[3] <https://arxiv.org/pdf/1802.03426>

[4] <https://www.tandfonline.com/doi/pdf/10.1080/00031305.2018.1527253>

[5] PMID: 31145670

[6] https://distill.pub/2016/misread-tsne/?_ga=2.135835192.888864733.1531353600-1779571267.1531353600